Luiz Cauet Castilla Ruy Blum

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EDUCATION

University of California, Santa Cruz — Santa Cruz, California
B.S. in Computer Engineering - Systems Programming, CS Minor — *Expected Graduation: March 2027*
GPA: 3.4
Related Coursework: Machine Learning, Data Structures & Algorithms, Networks, Systems Design, Computer Architecture, Software Engineering

EXPERIENCE

The Audition — Santa Cruz, California
Lead AI & Software Engineer — September 2025 – Current
- Classified guitar solo quality, achieving 85% agreement with human rankings by designing and training a PyTorch Convolutional Neural Network model.
- Defined high solo quality by collecting 768 pairwise listener judgments across 100 solo sequences through a comparison website and sequence database.
- Led software development while coordinating a 13-member multidisciplinary team.
- Architected gameplay systems in Godot, including a freestyle-performance mechanic where player input drives narrative.

Augmented Design Lab — Santa Cruz, California
Research Participant (Advisor: Dr. Jim Whitehead) — Jan 2026 – Mar 2026
- Built a reusable Skills feature for the Layered-Selection-Prompting research project, condensing a multi-prompt assisted-design workflow into a single prompt.
- Developed 4 procedural generation prototypes using LLMs (LangChain), Wave Function Collapse, and Z3 constraint solving.
- Built a Z3 constraint solver for tile-based map generation that guarantees connected floor layouts and entrance-to-exit paths.

Integem Inc. — Fremont, California
AI & Robotics Instructor — June 2026 – July 2026
- Taught robotics, computer vision, and generative AI to students ages 10–17.
- Guided 100+ students ages 10–17 through end-to-end YOLOv5 pipelines — dataset creation, annotation, training, tuning — to deployed vision models for robotic object detection and manipulation.
- Mentored students through NVIDIA's Generative AI curriculum, covering diffusion models and applied ML.

The Slug Collective — Santa Cruz, California
Founder & President — Dec 2024 – Current
- Founded and led a 100+ member interdisciplinary music and tech collective at UCSC
- Deployed an app with shared-taste discovery, private messaging, and event group generation.
- Secured $1,500 in funding through technical project proposals.

PROJECTS

Resonance - Semantic Preset Retrieval (Python, RAG, FastAPI, LLMs) — 2026
- Reduced preset discovery time by 60x, from 60+ seconds of manual browsing to ~1 second per query, by implementing CLAP-embedding vector-similarity search.
- Designed a multimodal retrieval-augmented generation (RAG) pipeline that maps natural-language descriptions to synthesizer presets, refining parameters iteratively from user feedback.
- Implemented Python + FastAPI backend services for embedding generation, preset retrieval, and parameter manipulation; deployed Dockerized ML pipelines on an Oracle Cloud VM.

Facial Expression Controller (Python, TensorFlow/Keras, OpenCV) — 2026
- Improved CNN validation accuracy by 25.4 percentage points (46.1% → 71.5%) through architecture and training experiments, while keeping the model to 141,675 parameters.
- Built an OpenCV webcam game controller that plays tic-tac-toe from facial expressions, with prediction smoothing, confidence gating, and explicit class-to-action mapping.

SmartQueue (Next.js, React, TypeScript, Supabase, Spotify SDK) — 2025
- Built a real-time Spotify web player whose weighted recommendation algorithm reorders queues from live skip, play, and replay signals, backed by Supabase Auth and row-level security, with 95th-percentile warm-load under 5 ms.

SKILLS

Languages: Python, C/C++, TypeScript, SQL, Java, Assembly
AI/ML: PyTorch, TensorFlow/Keras, CUDA, NumPy, SciPy, scikit-learn
Infrastructure: Linux, Docker, PostgreSQL, FastAPI, AWS, Wireshark